Nicolas Toon

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Education

B.S. in Data Science, *University of California, San Diego* — Sep. 2023 – Mar. 2027

- **GPA:** 3.94/4.0
- **Relevant Coursework:** Machine Learning Algorithms, Probabilistic Machine Learning, Data Visualization, Database Management, Data Structures & Algorithms, Probability/Calculus/Linear Algebra

Experience

Bioinformatics Analyst Intern, *Scripps Research* — Sep. 2024 – Present

- Writing Python functions with matplotlib and Plotly, synthesizing tabular data into time series visualizations
- Generating understandable deliverables reporting significant genetic mutations found using pandas and Dask
- Creating a full-stack application allowing researchers to visualize the mouse brain and co-expressed genes

Lead Camera Perception Engineer, *Triton AI* — Oct. 2025 – Present

- Training and deploying YOLO computer vision models with Roboflow, OpenCV, and Ultralytics to detect go-karts
- Utilizing Docker to test computer vision models and sensor fusion algorithms, increasing detection confidence
- Documenting and refactoring C++ and Python ROS2 software to streamline autonomous go-kart processes

UI Designer & Software Developer, *Data Science Student Society* — Oct. 2025 – Present

- Designing and creating websites using Figma, React, and Tailwind for San Diego organizations and events
- Adopting Agile methodologies to lead a team of undergraduates to ensure a quality product reaches clients

Projects

Paper Petals — Oct. 2025

- Designed a portfolio for a local creator to showcase their origami flower bouquets using Figma and React
- Automated commission form responses into a spreadsheet and email notifications using Apps Script
- Refactored TypeScript code to be compatible in mobile applications, increasing visibility and ease of interaction

Portfolio Website — Sep. 2025

- Built a responsive and mobile-friendly website that showcases my past projects using TypeScript and React

Stage Left: Typing with Parkinson's — Jun. 2025

- Collaborated with two undergraduates to develop an interactive and educational web tool for the general public
- Utilized D3 to create a custom typing game to race against a model, which simulated a Parkinson's patient

Classifying Benign and Malignant Breast Tumors — Jun. 2025

- Collaborated in a team of four students to research image segmentation capabilities of varying architectures
- Constructed a U-Net architecture model with a supervised dataset of 9000 images from scratch using PyTorch
- Integrated a binary classifier into the U-Net architecture, which achieved 90.2% recall

Recipe Representation — Mar. 2025

- Conducted exploratory data analysis and hypothesis tests to verify the significance behind online recipe tags
- Constructed a classifier with scikit-learn to make predictions about tags based on a given recipe's data

Skills

Languages: Python, SQL, JavaScript/TypeScript, Java, C++

Data Modeling: pandas, NumPy, SciPy, matplotlib, Plotly, Tableau, OpenCV, PyTorch, scikit-learn, Dask, D3

Tools: Git, Docker, ROS2, Excel, CVAT, Roboflow, React, Figma, Tailwind, Affinity, Photoshop, InDesign